Can a Quantum Circuit Read a Sentence?

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Motivation

We built a **Quantum Natural Language Processing (QNLP)** pipeline that classifies sentence sentiment as **positive** or **negative** using a parameterized quantum circuit. Grammar determines the circuit structure; each word contributes a learned rotation angle. The final measurement collapses the state to a class label. We train and evaluate on sentence pairs such as **the clever program runs** vs. **the broken code fails**.

positive $\equiv |1\rangle$ | **negative** $\equiv |0\rangle$

End-to-End Architecture

words $\rightarrow$ rotation parameters · grammar $\rightarrow$ circuit structure & weights · circuit $\rightarrow$ measure $\rightarrow$ class label

Each word is represented by a single trainable rotation angle. Grammar determines how those angles are composed into a circuit. Measuring the output qubit yields $P(|1\rangle)$, interpreted as the probability of **positive** sentiment.

Step 1 — Grammatical roles and learned weights

All sentences share a common syntactic structure; each word is assigned a colored **grammatical role**:

clever **program** **runs** **(the puzzle)**

adjective subject verb object

Grammar wires words together and weights each role's contribution. Verbs and adjectives carry more weight, since they drive sentiment:

role	weight
verb	**1.40**
adjective	**1.25**
object-adjective	0.90
subject	0.45
object	0.35

(Object-role weights only apply when the sentence has an explicit object; "the clever program runs" uses only the top three.)

Step 2 — Qubit states and measurement

A classical bit takes a definite value in $\{0, 1\}$. A **qubit** is a two-level quantum system whose pure state is a unit vector on the **Bloch sphere**. Before measurement, the qubit exists in a **superposition** of $|0\rangle$ and $|1\rangle$.

- An $R_y(\theta)$ rotation from $|0\rangle$ yields $|\psi\rangle = \cos(\theta/2)|0\rangle + \sin(\theta/2)|1\rangle$, a point at polar angle θ from the north pole of the Bloch sphere.
- **Measurement** in the $\{|0\rangle, |1\rangle\}$ basis projects $|\psi\rangle$ onto one of the basis states. The probability of each outcome equals the **squared magnitude of the projection**: $P(|1\rangle) = |\langle 1|\psi\rangle|^2 = \sin^2(\theta/2)$. The equatorial state ($\theta = \pi/2$) gives $P(|0\rangle) = P(|1\rangle) = \frac{1}{2}$.

We use a **2-qubit register**: a **subject qubit** and a **sentence qubit** whose measurement yields the classification. The joint state lives in the tensor product space $\mathcal{H}_{\text{subj}} \otimes \mathcal{H}_{\text{sent}}$; amplitudes of product states multiply.

Step 3 — Encoding words as rotation angles

Each word maps to an $R_y(\theta)$ rotation on its qubit wire. Grammar dictates which angles go on which wire. For **the clever program runs**, the *trained* parameters are **clever**$= 0.098$, **program**$= 0.043$, **runs**$= 0.489$:

$$\textbf{subject } \theta = \text{clever} + \text{program} = 0.098 + 0.043 = \mathbf{0.141}$$

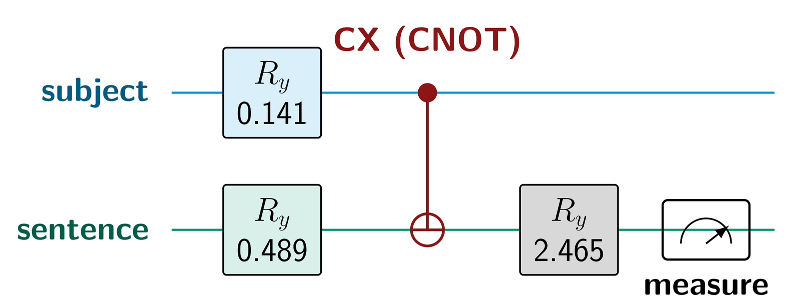
$$\textbf{sentence } \theta = \text{runs} = \mathbf{0.489}$$

$$\textbf{verdict angle} = \underbrace{\pi/2}_{\text{equatorial init.}} + \underbrace{\text{score}}_{\text{weighted sum}}$$

$$\text{score} = \underbrace{0.068}_{\text{bias}} + 1.25(0.098) + 0.45(0.043) + 1.40(0.489) = 0.894$$

$$\text{verdict angle} = \pi/2 + 0.894 = \mathbf{2.465}$$

The sentence qubit starts at the equatorial state ($\theta = \pi/2$), so the weighted score tips it toward either pole — the verb dominates (weight 1.40).



An $R_y(\theta)$ gate rotates the qubit state vector by angle θ about the y-axis of the Bloch sphere. The sentence qubit receives two successive R_y rotations: first by its word parameter (0.489), then by the verdict angle (2.465).

Step 4 — Entanglement via the CNOT Gate

Grammar requires the **subject** and **sentence** qubits to be treated as a single composite system. The **CNOT (CX) gate** encodes this grammatical composition:

If subject $= |1\rangle$: flip sentence qubit. If subject $= |0\rangle$: leave sentence qubit unchanged.

Action on the computational basis

The 2-qubit system has four computational basis states (left digit = subject, right = sentence). CNOT **swaps the two states where the subject is** $|1\rangle$ ($|10\rangle \leftrightarrow |11\rangle$):

	00	01	10	11
before	0.968	0.241	0.068	0.017
after	0.968	0.241	0.017	0.068

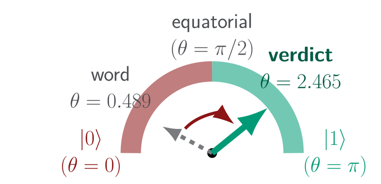
Because each qubit is in superposition, the CNOT produces an **entangled** state: the result can no longer be written as a tensor product $|\psi_{\text{subj}}\rangle \otimes |\psi_{\text{sent}}\rangle$ of the two qubits separately.

Forward Pass: Full State Evolution

The 2-qubit state is tracked as four computational basis amplitudes ($|00\rangle, |01\rangle, |10\rangle, |11\rangle$; **left digit** = **subject**, **right** = **sentence**). Each gate reshuffles these amplitudes.

operation	00	01	10	11
initialized to $\lvert 00\rangle$	1.000	0.000	0.000	0.000
apply word-parameter R_y to each qubit	0.968	0.241	0.068	0.017
CNOT: $\lvert 10\rangle \leftrightarrow \lvert 11\rangle$	0.968	0.241	0.017	0.068
apply verdict $R_y(2.465)$ to sentence qubit	0.094	0.993	-0.059	0.039

The sentence qubit's word angle ($\theta = 0.489$) leaves it near the $|0\rangle$ pole; the verdict ($\theta = 2.465$) pushes it past the equator into the $|1\rangle$ hemisphere:



Measuring the sentence qubit: outcomes $|01\rangle$ and $|11\rangle$ (right digit = 1, **green** columns) are positive. The probability of each outcome is the squared amplitude:

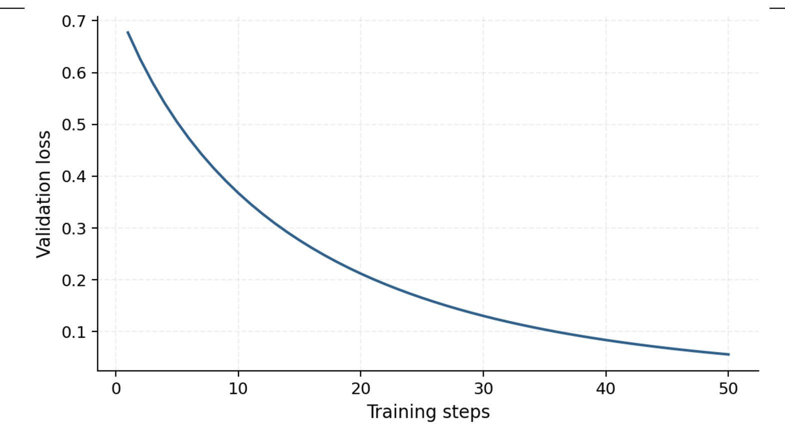
$$P(|1\rangle) = 0.993^2 + 0.039^2 = \mathbf{0.988} \Rightarrow \textbf{POSITIVE} ✓$$

The negative counterpart **the broken code fails** yields $P(|1\rangle) = \mathbf{0.003} \Rightarrow$ **NEGATIVE**.

Classification Results

	Correct
Training (60)	**100%**
Test (12)	**100%**

All predictions come from exact statevector simulation. At **2 qubits** this runs in < 1 s; scaling to real language would require many entangled qubits and actual quantum hardware.



Training: the parameter-shift rule

Error: $e = P(|1\rangle) - y_{\text{label}}$. Gradients are computed exactly by re-running the circuit at $\theta \pm \frac{\pi}{2}$:

$$\frac{\partial P}{\partial \theta} = \tfrac{1}{2}\left[\, P(\theta + \tfrac{\pi}{2}) - P(\theta - \tfrac{\pi}{2}) \,\right], \qquad \theta \leftarrow \theta - \eta \left(e \cdot \frac{\partial P}{\partial \theta} \right)$$

This identity holds exactly for gates $e^{-i\theta G/2}$, making it executable on real quantum hardware. Training proceeds for ~ 50 epochs.

References

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